Verigate PortalDot

Authorization middleware for AI-agent payments on PortalDot V3

A proposal is not authority.

Verigate evaluates the action. The bridge verifies signed evidence. A dedicated Solidity guard enforces the exact approved transfer.

VERIGATE	BRIDGE	PORTALDOT
Ed25519 decision	Exact action binding	EIP-712 guard

TESTNET

420420777

PortalDot V3 EVM testnet

Contract deployed • Receipt verified

Track fit

Modular Cloud Solutions and Middleware

Built around a narrow, reusable control point between AI intent and token execution.

Preventing unrestricted agent spending

The authorization must bind the decision to the exact thing the executor is allowed to do.

01 RECIPIENT SUBSTITUTION

An approved payment must not be reused with a different destination.

02 AMOUNT TAMPERING

The amount must remain the same from signed intent to contract call.

03 REPLAY OR EXPIRY

A valid authorization should not be usable again or after its deadline.

Design principle

A signed ALLOW is necessary but not enough: its authority must be short-lived, narrowly scoped, and bound to a recipient, amount, chain and contract.

A narrow path from decision to execution

Off-chain verification and on-chain enforcement have separate, explicit trust boundaries.

1

AGENT INTENT

Proposed action

2

VERIGATE

Signed ALLOW + execution authorization

3

BRIDGE

Verify Ed25519; bind exact action

4

EIP-712

Chain, guard, recipient, amount, nonce, deadline

5

GUARD

Reject invalid/replayed or altered transfer

WARN / BLOCK / invalid signature / mismatched action

The bridge refuses to sign; the contract cannot independently override this off-chain decision.

Why PortalDot V3 — and where this project fits

The project uses PortalDot's execution environment; it does not overstate a cross-chain bridge that is not implemented.

- **EVM-compatible contract execution**

  The guard is written in Solidity and uses EIP-712 signatures bound to chain ID and contract address.

- **Substrate-aware operational checks**

  The tooling separately checks the configured Substrate WebSocket endpoint and genesis hash before sensitive network operations.

- **Reusable middleware contribution**

  Other agent-enabled applications can reuse a strict authorization boundary instead of giving an agent a direct, unrestricted transfer path.

TRACK

Modular Cloud Solutions and Middleware

Not claiming:

A deployed multi-chain asset bridge, verified wallet mapping, production readiness, or an independent audit.

PortalDot V3 testnet deployment evidence

The contract deployment is a real testnet interaction; the full transfer workflow has not yet been shown on the public network.

NETWORK

PortalDot V3 EVM testnet

CHAIN ID

420420777

CONTRACT

0x3f2f4844...F09655b

DEPLOYMENT TX

0xb1cf9cca...1fcdce3c

RECEIPT

Success • block 116872

RUNTIME CODE HASH

0xb3d71ff8...f7f7671

READ-BACK

2,153 bytes

Runtime bytecode

authoritySigner()

On-chain value matched the configured signer.

Console

node-console.portaldot.dev

What has been verified — and what has not

Tests and local execution evidence are labelled separately from public testnet deployment evidence.

156
local tests passed

ALLOW
valid signed artifacts

BLOCK
execution refused

- **Dry-run did not change state**
  The executor simulated the exact authorized transfer before execution.

- **Local EVM transfer matched authorization**
  TransferExecuted was confirmed and recipient balance delta matched 1,000,000 base units.

IMPORTANT DISTINCTION

The end-to-end transfer executed only on a temporary loopback Hardhat EVM configured with the target chain ID. No public PortalDot transfer was sent.

Next steps and ecosystem value

The immediate work is reproducibility and honest evidence, not another deployment.

- **Publish the current implementation branch**

  Preserve the event-period commit history and run CI on the exact public head.

- **Record a short product demo**

  Show the actual ALLOW/BLOCK path, local simulation, and the public testnet deployment evidence as distinct steps.

- **Close operational gaps**

  Verify the specific H160-to-SS58 mapping before using that wallet pair and plan any live transfer with explicit human approval.

ECOSYSTEM VALUE

A reusable spending-control component, a minimal guard contract, deployment evidence, fail-closed network tooling and an operations runbook.

LIMITATIONS

Testnet prototype. No independent security audit. Specific EVM/SS58 mapping unverified. Public end-to-end transfer demo still pending.